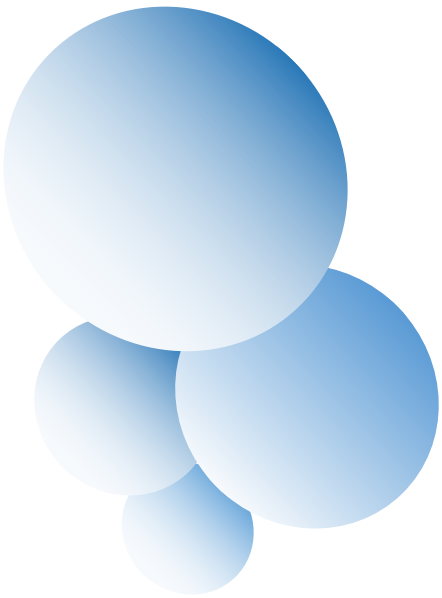
EBA5002: Graduate Certificate in Business Analytics Practice

Multi-Dimensional Approach to Estimating Employee Burnout

HumanFirstCollective_Final_Report.pdf



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HumanFirst Collective

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Contributions

Module / Task	Name	Contribution
Anomaly Detection & Association Rules	Bhavaraju Anuraag	100%
Dimensionality Reduction & Clustering	Bhavaraju Anuraag	100%
Text Analysis - Sentiment Analysis & Topic Modelling	Yangyi Zhang	100%
Predictive Modelling using Tree-Based Models	Charulata Chouhan	100%
Predictive Modelling - Sliding Bidirectional Recurrent Neural Network	Indrajit Arun	100%
Predictive Modelling - Transformers	Gayathri	100%
Key Insights (Tableau Dashboards)	Yangyi Zhang	100%

2. Introduction and Background

Employee burnout is one of the most significant challenges faced by modern organizations, directly impacting productivity, engagement, innovation, and retention. World Health Organization has included burnout in the 11th Revision of the International Classification of Diseases (ICD-11) as an occupational phenomenon (WHO, 2019).

Employees who are suffering from burnout are more likely to experience sleep disturbances, cardiovascular diseases , depression amongst many other health concerns Gallup estimates that low employee engagement (a hallmark of burnout) costs the global economy $8.8 trillion, or 9% of the global gross domestic product (Gallup, 2021; Wiens, 2024).

Hence organizations across sectors such as technology, consulting, finance, healthcare, and education are increasingly leveraging data science to understand workforce dynamics. Burnout has become a data-critical HR issue due to the different reasons mentioned above. This project aims to enable organizations to detect early warning signs, implement preventive strategies, and foster healthier, more sustainable work environments.

3. Problem Statement

The contemporary workforce faces significant, quantifiable risks associated with employee burnout, which directly translates into critical business challenges across organizations. The core problems this project seeks to address are defined by the impact, timeliness, and complexity of burnout identification.

Attrition: The most immediate and costly consequence of unmanaged burnout is attrition. High turnover rates lead to significant financial burdens associated with recruiting, training, and knowledge loss. Furthermore, the presence of burnout drives higher rates of absenteeism, severely damages team morale, and ultimately diminishes overall organizational productivity.

Lack of Early Detection: A critical gap in current organizational strategy is the lack of an early-warning system to identify the onset of employee burnout. Most interventions occur reactively, after an employee's productivity has already declined or they have made the decision to leave. The business requires a proactive, data-driven system to flag at-risk individuals before burnout reaches a critical stage, thereby enabling timely and effective preventive measures.

Complexity: Employee burnout is not a simple, single-variable issue; it is a complex psychological phenomenon defined by feelings of Exhaustion**,** Depersonalisation, and reduced Personal Growth. Traditional HR metrics often fail to capture the nuance of these underlying psychological factors. This complexity makes accurate, timely identification challenging and necessitates advanced analytical techniques to transform subjective feedback into objective, measurable, and actionable latent dimensions.

4. Business Objectives

To effectively address the business problem of employee burnout and its associated costs, this project is guided by a set of clear business objectives. These focus on the high-level organizational impact and the delivery of actionable intelligence.

Identify Burnout Drivers: Quantify and measure the key organizational, behavioural, and psychological factors (latent variables) that most significantly influence employee burnout. This involves dimensionality reduction and latent factor analysis.
Reduce Attrition: Utilize the predictive and segmented insights generated by the model to prevent employee exits that are demonstrably linked to high burnout or low satisfaction scores.
Improve Engagement & Morale: By identifying and targeting the specific factors driving negative sentiment (e.g., poor work-life balance), the project aims to foster more positive work environments, thereby improving overall job satisfaction and engagement.
Support HR Strategy: Enable the Human Resources department to make timely, evidence-based decisions regarding workforce well-being, resource allocation, and program effectiveness based on quantitative risk profiles rather than anecdotal evidence.
Enhance Productivity & Morale: Create a mechanism for the early detection of at-risk employee segments to allow for preventive care and targeted intervention, ensuring that productivity and morale are maintained.
Provide Visual Decision Tools: Deliver interpretable results in the form of clear visualizations and segmented clusters, providing decision-makers with intuitive dashboards that summarize burnout risk, sentiment trends, and key performance metrics.

5. Technical Objectives

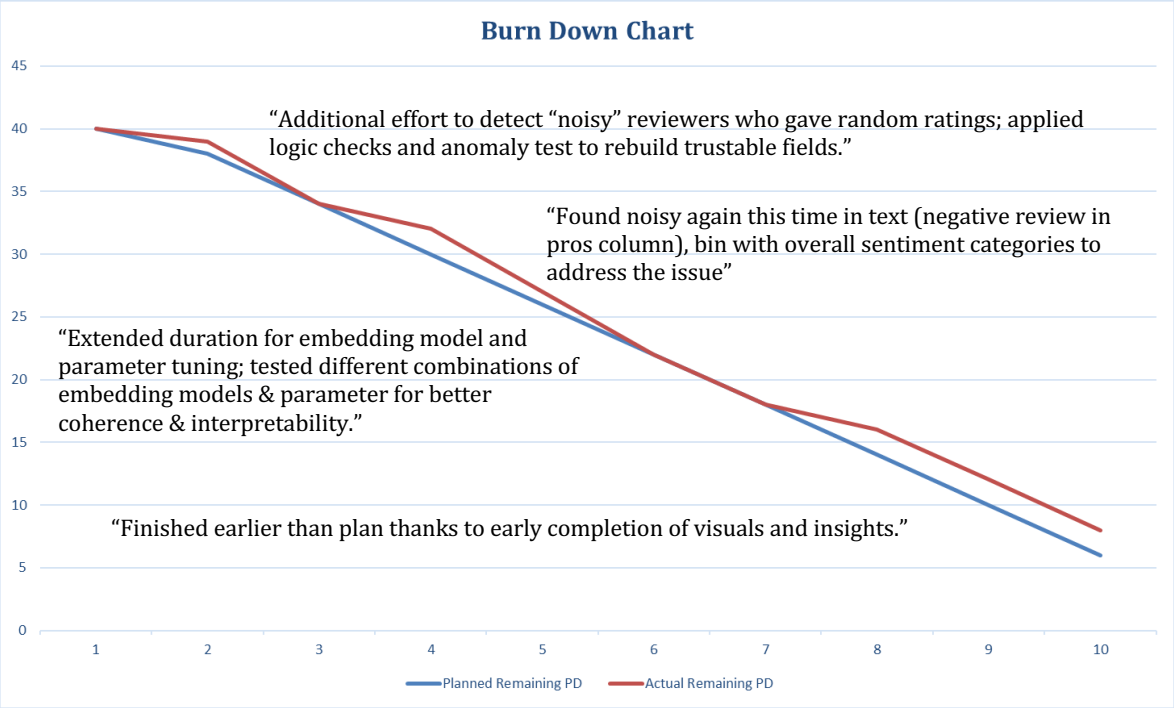
The technical objectives of this project are designed to translate the complex business problem of employee burnout into actionable, quantitative insights. This involves a multi-stage data science pipeline focused on data consolidation, feature engineering, and machine learning.

Technical Objective	Description	Business Alignment
Data Integration & Preprocessing	Collect and clean structured (HR, survey) and unstructured (feedback text) data.	Ensures a unified and reliable dataset to identify accurate burnout indicators.
Predictive Modelling	Use machine learning models to predict burnout risk levels and identify top influencing factors.	Enables early detection of high-risk employees and informs timely HR interventions.
Text Analysis (NLP)	Perform sentiment analysis, topic modelling, and keyword extraction from employee feedback.	Captures emotional and behavioural cues, offering deeper insights beyond numeric scores.
Visualization Dashboards	Develop interactive dashboards to display burnout risk, key drivers, and sentiment trends.	Translates analytical findings into actionable insights for HR and management teams.

6. Project Design

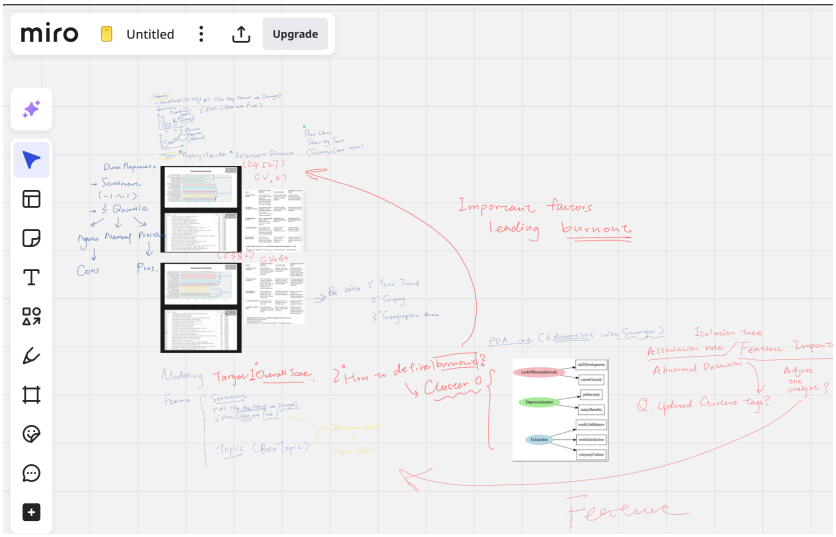
Effective project planning and agile execution were critical to successfully navigating the technical complexities of this initiative. Our work was structured using a sprint-based approach, underpinned by rigorous tracking and collaborative tools, ensuring both transparency and momentum.

Project Check-ins & Tracking: Project progress was continuously monitored using a Burn Down Chart to track weekly effort and the remaining work against the planned timeline. This visual tool provided immediate feedback, allowing us to finish key components ahead of the original schedule and maintain a consistent velocity. Daily 10-15 minute check-ins addressed immediate blockers and updated next steps, while weekly sprint reviews were used to formally review deliverables and adjust the workload.



Clear Accountability: We utilized a shared workflow that assigned clear ownership for key processes: Data Preparation, Exploratory Data Analysis (EDA), Modelling, Visualization, and Reporting. Roles were occasionally rotated to ensure broad skill development across modules.

Tooling and Traceability: Our collaboration relied on Teams for real-time communication, Miro for visual planning and initial brainstorming, and version-controlled updates to maintain a transparent and efficient pipeline. All code, data files, and slides were archived for full traceability.



7. Scope of Work

Analytical Techniques

Decision Trees & Random Forests **:** Can be used to identify key factors affecting employee satisfaction and company reputation.
Dimensionality Reduction**:** Techniques like Principal Component Analysis (PCA) simplify datasets by projecting features onto latent components, making predictive modelling and visualization more tractable.

Factor Analysis: This technique will identify latent factors underlying multiple observed variables related to employee perceptions and ratings.

Sentiment Analysis: Applied on textual data to assess and understand the overall employees' reviews and comments.

Clustering Analysis: Clustering algorithms (e.g., K-means, hierarchical clustering) can group employees into three segments—Burnout, Transition to Burnout, and Healthy—which allows targeted intervention strategies.

Topic Modelling: Latent Dirichlet Allocation (LDA) can be used on textual data to discover prevalent themes that employees mention. This technique uncovers important areas of satisfaction and concern that organizations can address proactively.

Evaluation Methodologies

- **Classification:** Recall, AUC-ROC
- **Forecasting:** RMSE, MAE, MAPE, R^2
- **Internal Cluster Validation:** Metrics like silhouette score and Davies-Bouldin index will evaluate cluster cohesion and separation.
- **Topic Coherence Measures:** Use quantitative coherence scores (e.g., UMass, CV) for optimal number of topics and meaningful themes.

8. Dataset Used & Data Description

Data Source: AmbitionBox, a career advisory company, is India's largest platform for company reviews, salary insights, and interview questions (**Link:** https://www.ambitionbox.com)

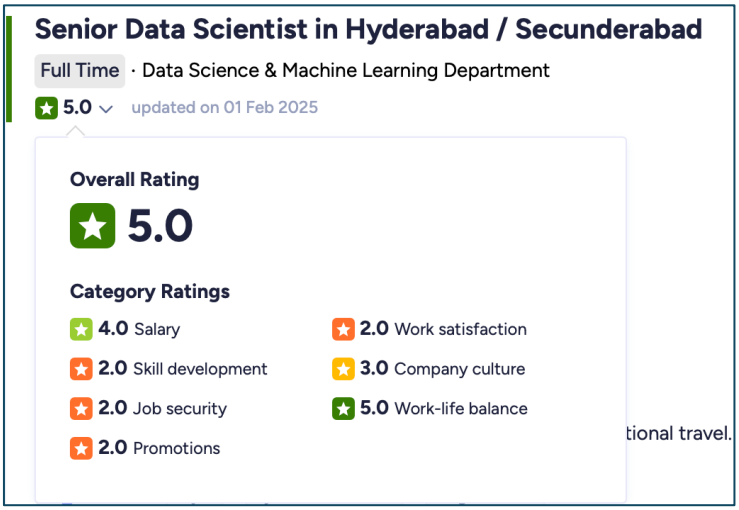
Data Collection Methodology:
The dataset was collected through web-scraping tools designed to systematically extract publicly available information from the AmbitionBox website, primarily utilizing Python libraries like Scrapy to automate the scraping and data extraction processes.

Data Dictionary:

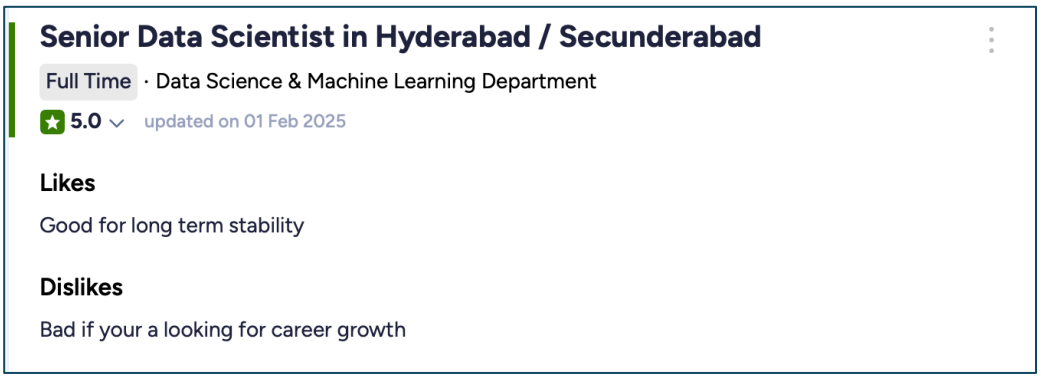
Variable Name	Description	Data Type
EmployeeIndex	Unique identifier for each employee review entry	Integer
jobFunction	Job function or role title	Text
company	Name of the company	Text
location	Geographic location of the job	Text
salaryBenifits	Rating on salary, benefits	Integer
workSatisfaction	Rating on work satisfaction	Integer
skillDevelopment	Rating on skill development opportunities	Integer
companyCulture	Rating on company culture	Integer
jobSecurity	Rating on job security	Integer
worklifeBalance	Rating on work-life balance	Integer
careerGrowth	Rating on career growth opportunities	Integer
likes	Positive aspects above the job/company	Text
dislikes	Negative aspects about the job/company	Text
reviewText	Textual review or comments about the job/company	Text
postedOn	Date when the review was posted	Date

9. Data Understanding

Each review includes both quantitative ratings—such as salary, work-life balance etc.,

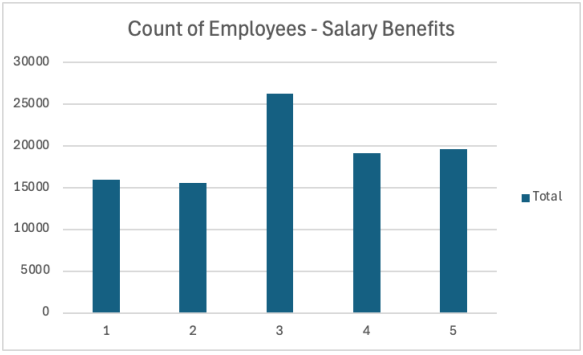


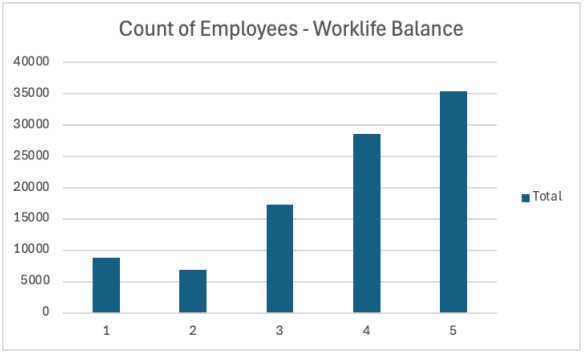
And textual feedback describing employees' likes and dislikes about their workplace



10. Exploratory Data Analysis

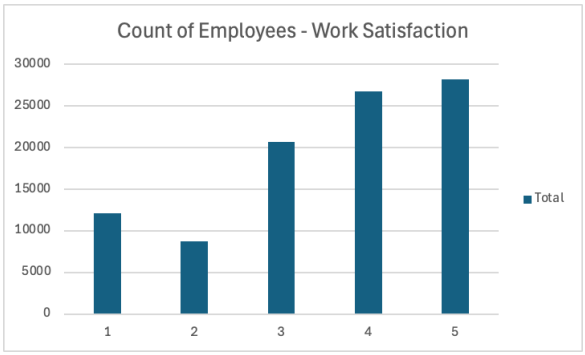
The following analysis details the frequency distributions of seven employee satisfaction variables, measured on an ordinal scale from 1(Least Satisfied) to 5 (Most Satisfied). The characterization of each distribution focuses on its shape, mode, and central tendency.

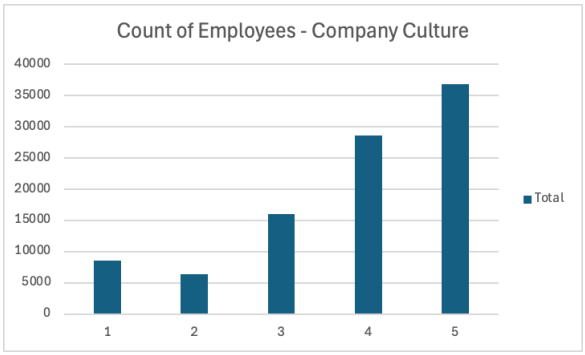




Salary Benefits: This variable exhibits a centralized tendency (approaching normal distribution). The mode is located at the neutral rating 3. Frequencies for the highest and lowest classes are lower but comparable. This profile suggests a mean and median close to the central point of the scale.

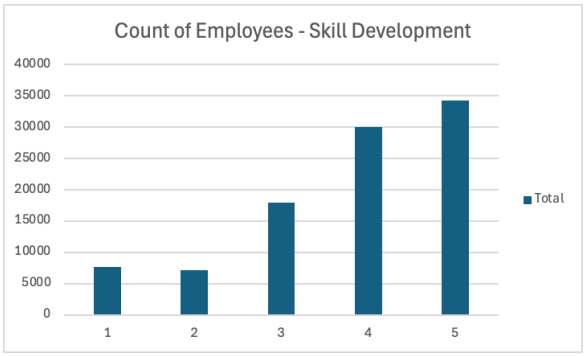
Work-Life Balance: Work-Life Balance presents a distribution with a strong positive skew. The majority of observations are pooled in the upper classes. The lowest ratings constitute the smallest subsets of the population.

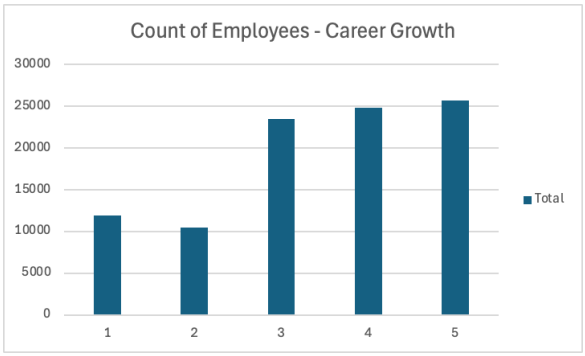




Work Satisfaction: This distribution is positively skewed. Frequencies are highest at r= 4 and 5 Critically, the frequency at the neutral rating is substantial, indicating the distribution's mean is less displaced toward the maximum rating.

Company Culture: The distribution shows a strong positive skew. The modal rating is r=5. Frequencies decline steadily as satisfaction decreases.

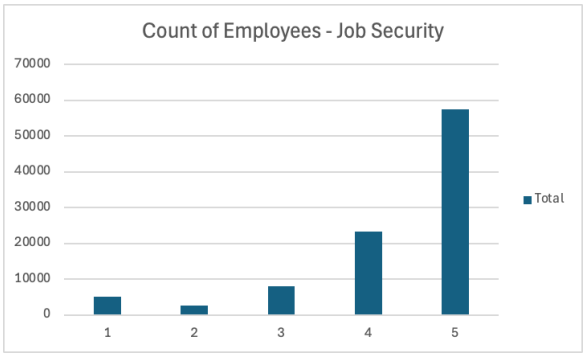




Skill Development

This metric displays a strong positive skew. The sum of frequencies for r in (4, 5) substantially outweighs the combined frequencies for r in (1,2,3)

Career Growth: This distribution approaches a mesokurtic shape on the upper half of the scale. Frequencies are relatively uniform across the upper three rating classes. The highest frequencies are concentrated in these three adjacent classes.



Job Security

This variable exhibits the highest degree of positive skewness among all metrics. The distribution is highly concentrated at r=5. Frequencies decline sharply toward the lower ratings

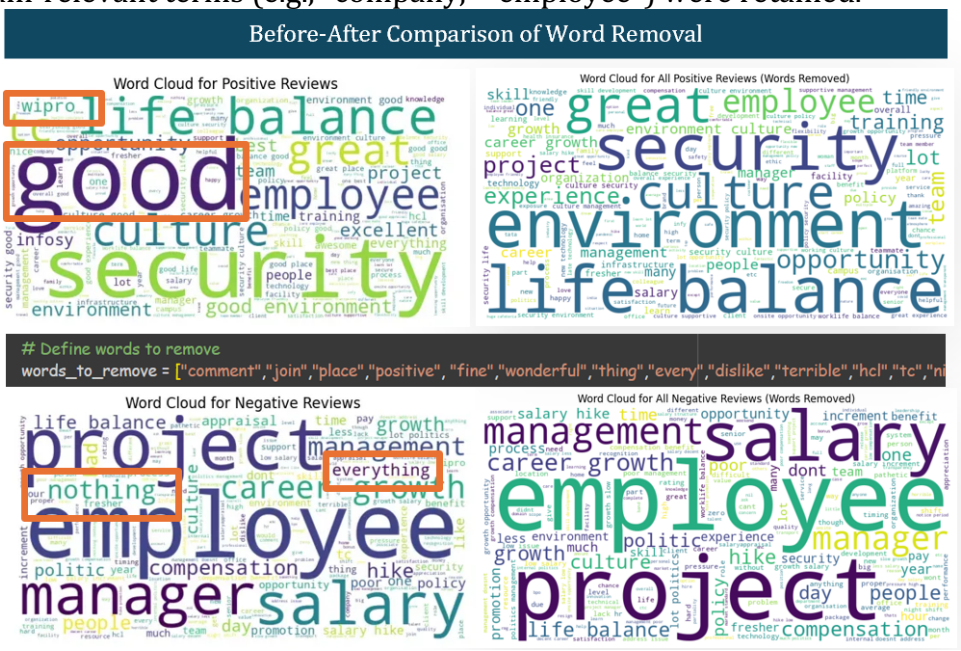
11. Data Preparation & Transformations

Missing Value Handling

A preliminary inspection of the feature set revealed that all seven primary variables were initially processed to remove any instances containing null or missing values in their respective numeric columns. This measure is to eliminate the need for imputation and maintaining the integrity of the original frequency counts. No further transformations (e.g., scaling or encoding) were required as all features are ordinal and on the same 1-5 scale.

Text Preprocessing for Topic Modelling

1. Text Consolidation and Cleaning: Multiple text fields (e.g., pros, cons, comments) from each review were merged into a single document to preserve full context. Irrelevant content, including URLs, digits, and extraneous punctuation, was removed.
2. Filtering and Retention: Common stopwords were eliminated, while specific domain-relevant terms (e.g., "company," "employee") were retained.



3. Part-of-Speech (POS) Filtering: A critical step involved retaining only nouns and adjectives. This ensures the focus is on semantically rich tokens—nouns representing key aspects (manager, policy), and adjectives conveying sentiment or quality (toxic, supportive). Filtering out other POS categories (like verbs) enhances topic coherence and interpretability by eliminating irrelevant filler words.
4. Lemmatization: Remaining tokens were lemmatized to their base form (e.g., *learning* to *learn*). This process reduces data sparsity and unifies word variants, a necessary step for consistent quantitative analysis.

12. Analysis & Modelling Performed & Evaluation Strategies

Anomaly Detection: Isolation Forest

This phase of the analysis focused on identifying non-conforming patterns within the employee review data that could potentially bias predictive modelling or represent actionable, isolated feedback cases.

Data Preparation and Feature Engineering

The anomaly detection process utilized seven ordinal satisfaction features. Feature engineering consisted of Standard Scaling the data. Standardization transforms the features to have a mean of 0 and a standard deviation of 1, which is a standard prerequisite for distance-based or distribution-sensitive algorithms like Isolation Forest, aiding in the fair identification of records regardless of their original value range.

Model and Hyperparameter Tuning

The Isolation Forest (IF) algorithm was selected for anomaly detection. IF is an ensemble tree-based method that isolates anomalies (outliers) rather than profiling normal data points. It works by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature. Anomalies, being few and far from the rest of the data, are typically separated from the normal observations closer to the root of the tree, requiring fewer splits.

The model was fine-tuned using the contamination hyperparameter, set explicitly to 0.05. The model was instructed to identify approximately 5% of the total records as anomalies, providing a targeted and constrained approach to outlier removal consistent with standard data cleaning practices.

Model Inference

The Isolation Forest successfully detected a distinct set of anomalous employee reviews**,** identifying 5% of the records as outliers based on their low anomaly score and low decision function value**.**

The most critical insight is the identification of records exhibiting inconsistent response patterns—for instance, an employee providing all minimum ratings across individual satisfaction metrics but assigning a relatively high overallScore. This inverse pattern strongly suggests:

1. **Data Quality Issues:** Potential survey error, misinterpretation of the scale, or data entry corruption.
2. **Unique Respondent Behavior:** An unusual psychological profile or deliberate noise injection.

	skillDevelopment	salaryBenefits	workLifeBalance	workSatisfaction	jobSecurity	careerGrowth	companyCulture	overallScore	anomaly_score	anomaly_value
86798	5.0	1.0	1.0	1.0	1.0	1.0	1.0	4.0	-1	-0.184391
5395	1.0	1.0	1.0	1.0	1.0	1.0	1.0	4.0	-1	-0.180723
31815	1.0	1.0	1.0	1.0	1.0	1.0	1.0	4.0	-1	-0.180723
80325	3.0	1.0	1.0	1.0	1.0	1.0	1.0	4.0	-1	-0.176031
86666	1.0	1.0	1.0	1.0	2.0	1.0	1.0	4.0	-1	-0.174474
62312	4.0	1.0	1.0	1.0	1.0	1.0	1.0	4.0	-1	-0.172920
83351	1.0	1.0	1.0	5.0	5.0	1.0	1.0	4.0	-1	-0.168276
82793	1.0	3.0	1.0	1.0	1.0	1.0	1.0	4.0	-1	-0.168108
92041	1.0	1.0	1.0	1.0	1.0	1.0	2.0	4.0	-1	-0.165708
10058	5.0	1.0	2.0	1.0	1.0	1.0	1.0	4.0	-1	-0.165196

The model successfully highlighted records with statistically unusual feature combinations against the overall score. These anomalous records can now be exempted from the core predictive modelling scope, preventing them from distorting the feature weights and improving the overall robustness and generalizability of future models. This ensures the predictive model focuses only on consistent, "normal" feedback patterns.

Association Rules: Apriori Algorithm

This analysis utilized the Apriori algorithm to uncover frequent item sets and strong association rules, focusing on the relationship between ratings across individual workplace factors and the overallScore. This technique aims to identify structural dependencies within employee feedback.

Data Preparation and Feature Engineering

The seven ordinal satisfaction features and the overallScore were first one-hot encoded. Since the input for the Apriori algorithm must be a transactional, binary format, each rating in {1, 2, 3, 4, 5} for every column was converted into a distinct boolean feature (e.g., textsalaryBenefits_1, salaryBenefits_2, ...,overallScore_5). This process transformed the numeric ordinal data into a sparse, binary matrix suitable for frequent pattern mining.

Model and Hyperparameter Tuning

The Apriori algorithm was employed to efficiently generate frequent item sets. An itemset is considered "frequent" if its Support—the fraction of transactions (employee reviews) containing the itemset—exceeds a predefined threshold.

The primary hyperparameters used were:

1. min_support: Set to 0.10. This threshold ensured that only item sets present in at least 10% of all employee reviews were considered. This filters out rare, statistically insignificant patterns, focusing the rule generation on broadly applicable trends.

2. rule_metric: Set to lift with a min_threshold of 1.0. Lift measures how much more likely the consequent is, given the antecedent, than if the antecedent and consequent were independent. A lift greater than 1.0 indicates a positive and significant association.

Model Evaluation and Inference

antecedents	consequents	support	confidence	lift
workSatisfaction_1.0	overallScore_1.0	0.883529	1.0	1.0
salaryBenefits_1.0	overallScore_1.0	0.837172	1.0	1.0
companyCulture_1.0	overallScore_1.0	0.776184	1.0	1.0
workLifeBalance_1.0	overallScore_1.0	0.742865	1.0	1.0
careerGrowth_1.0, skillDevelopment_1.0	overallScore_1.0	0.683616	1.0	1.0

The most significant rules, filtered to target low overallScore as the consequent, revealed an exceptionally strong predictive pattern. The top rules, such as workSatisfaction_1.0 → overallScore_1.0 and salaryBenefits_1.0 → overallScore_1.0, demonstrate:

- High Support: The antecedent and consequent co-occur frequently (e.g., 88.35% for workSatisfaction_1.0).
- Perfect Confidence: The Confidence for all observed rules is 1.0. This implies that in every single instance where an employee assigned the minimum rating (r=1) to a core satisfaction factor, they also assigned the minimum rating (r=1) to the overallScore.

The Apriori analysis provided crucial evidence of the high predictive power of individual features. It reveals that a single rating can be a deterministic indicator of a low overall score. These insights are essential for subsequent predictive modelling, confirming that the lowest-rated features are not just important, but are primary drivers of minimum overall satisfaction.

Dimensionality Reduction: Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was performed to achieve dimensionality reduction, simplifying the multi-dimensional structure of the employee review data while preserving the maximum possible variance. This technique is crucial for visualizing the underlying patterns and verifying the separability of data points before applying clustering or classification algorithms.

Data Preparation and Feature Engineering

Prior to decomposition, the data underwent Standard Scaling to ensure that all features contributed equally to the principal components, regardless of their original scale or

spread. This step is necessary to prevent features with larger variance from disproportionately dominating the projection axes.

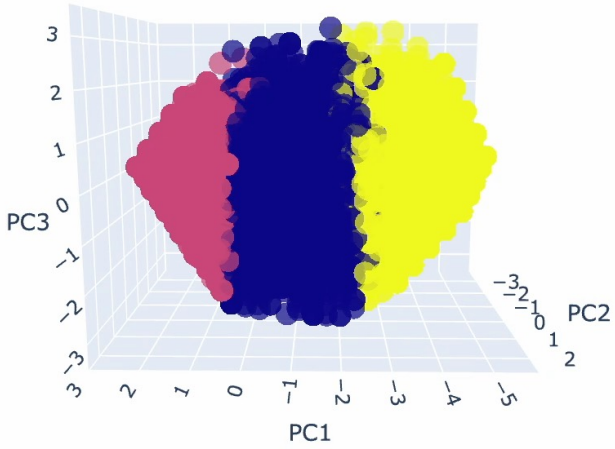
Model and Fine-Tuning

The PCA model was configured to extract three principal components n_components=3. This configuration compresses the 7-dimensional feature space into a 3-dimensional space, specifically chosen to allow for the clear visualization of the projected data. The components (PC1, PC2, PC3) are orthogonal axes that capture the directions of maximum variance in the data.

Model Inference

The projection of the scaled data into the 3-dimensional PCA space revealed meaningful separation, as illustrated by the clustering structure .

- PC1 vs PC2 & PC1 vs PC3 Axes: The resulting clusters exhibit distinct separation when viewed across PC1 in combination with PC2 and PC3. This clear demarcation along the primary axes suggests that the original 7 variables contain strong, correlated patterns that effectively group employee feedback into distinct perceptual segments
- PC2 vs PC3 Axis: The separation is observed to be less distinct or more diffused along the plane defined by PC2 and PC3, indicating that the majority of the distinguishing variance is successfully captured by the first principal component (PC1).



PCA successfully reduced the feature space while confirming the presence of inherent clustering structures within the employee review data. This result validates that the 7 features are not random, but collectively define separable groups of employees, which is a key prerequisite for effective segmentation using unsupervised learning techniques like K-Means clustering.

Dimensionality Reduction: Latent Factor Analysis

Following Principal Component Analysis (PCA), which identified underlying statistical structures, Latent Factor Analysis (CFA) was employed for a theory-driven approach to dimensionality reduction. The objective was to validate the alignment of the 7 observed satisfaction variables with three predefined psychological latent constructs, moving beyond purely statistical variance to focus on model interpretability.

Model Description and Construction

Factor Analysis is a Structural Equation Modelling technique used to test whether the relationships between a set of observed variables (the employee ratings) and their underlying latent factors (the psychological dimensions) conform to a predetermined theoretical model. The model structure was defined based on established occupational psychology literature, mapping the seven features to three latent factors:

$$\begin{aligned} \text{Exhaustion} &\approx \text{salaryBenefits} + \text{workLifeBalance} \\ \text{Depersonalisation} &\approx \text{companyCulture} + \text{workSatisfaction} + \text{jobSecurity} \\ \text{PersonalGrowth} &\approx \text{careerGrowth} + \text{skillDevelopment} \end{aligned}$$

Model and Output

The model was fit to the ordinal data, yielding factor loadings that quantify the strength of the relationship between each observed variable and its designated latent factor. The factor loading results (filtered to show only latent-to-observed relationships) confirm that all observed variables significantly contribute to their hypothesized latent factor.

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	salaryBenefits	~	Exhaustion	1.000000	-	-	-
1	workLifeBalance	~	Exhaustion	0.958608	0.003639	263.398277	0.0
2	companyCulture	~	Depersonalisation	1.000000	-	-	-
3	workSatisfaction	~	Depersonalisation	1.129072	0.002981	378.787363	0.0
4	jobSecurity	~	Depersonalisation	0.643866	0.002959	217.630979	0.0
5	careerGrowth	~	PersonalGrowth	1.000000	-	-	-
6	skillDevelopment	~	PersonalGrowth	0.867039	0.002537	341.819557	0.0

K-Means Clustering

Model and Fine-Tuning

The final stage of the analysis involved segmenting the workforce by applying the K-Means Clustering algorithm to the three theoretically derived latent factors Exhaustion, Depersonalisation, PersonalGrowth. This step moves beyond merely understanding data structure to grouping employees based on shared psychological experiences.
The model was trained exclusively on the factor scores resulting from the Confirmatory Factor Analysis (CFA). This ensures that the clusters are defined by the theoretically meaningful dimensions, rather than the raw, correlated ordinal ratings.

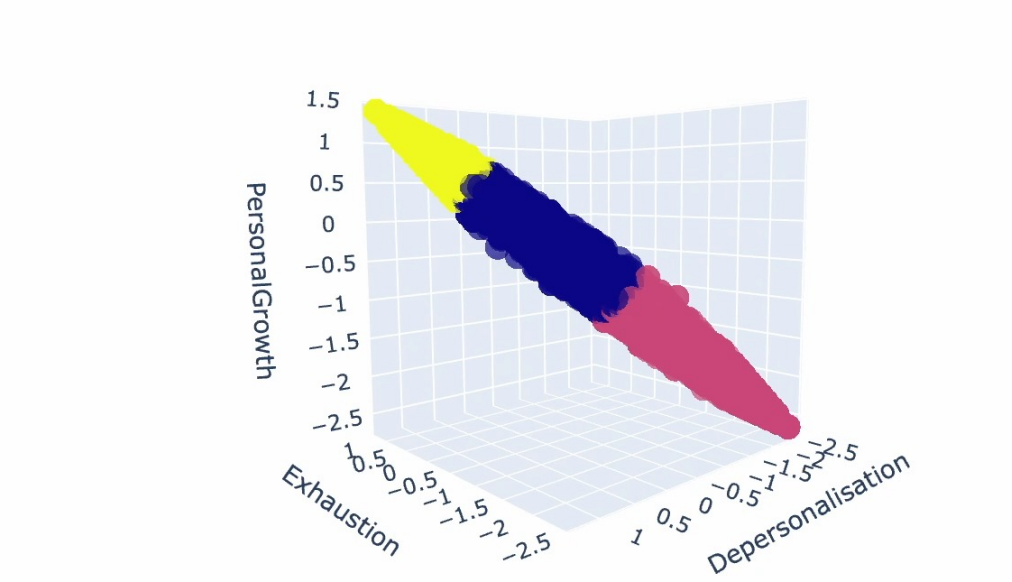
The K-Means algorithm was selected due to its efficiency and the visually distinct, non-overlapping nature of the clusters observed in the 3D latent space. The number of clusters was set to k=3}, a choice supported by visual inspection of the 3D factor score plot, which showed clear separation among three primary groups – Healthy, Transition to Burnout and Burnout clusters

Model Inference

The clustering successfully partitioned the data into three distinct, non-arbitrary segments, which can be interpreted based on their position in the latent factor space.
Distinctiveness: The clusters are highly distinct in the 3D space, indicating that employees fall into clearly separable groups based on their collective ratings across the three psychological factors.

This enables the transformation of continuous employee feedback into discrete, actionable workforce segments. This serves as a strong technical foundation for the business objective by allowing the organization to:

- Target Interventions: Identify and analyze the demographic and operational characteristics of each segment (e.g., highly exhausted vs. high depersonalisation).
- Predict Behaviour: Use the cluster ID as a new categorical feature in future supervised models aimed at predicting outcomes like employee turnover or performance.

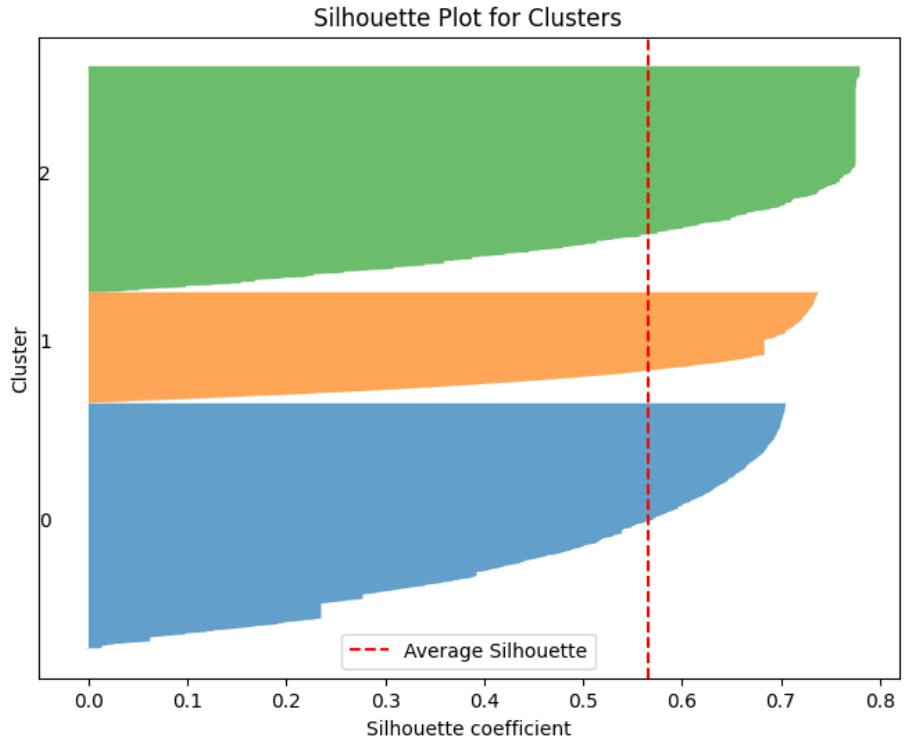


When the employee review data is projected onto the three latent factors—Exhaustion, Depersonalisation, and Personal Growth—the resulting visualization shows distinct clustering across all three axes.

Unlike PCA, where clusters are defined purely by statistical variance, the CFA visualization confirms that the defined latent constructs represent meaningful, independent, and well-separated psychological dimensions of the employee experience. This confirms the viability of using these three latent factors in subsequent modelling stages, providing two primary technical advantages:
Interpretability: The factors provide a direct, actionable link to established psychological concepts, such as those related to burnout (Exhaustion and Depersonalisation), enabling more targeted organizational interventions.
Robust Feature Set: By validating these three theoretical factors, the model ensures the input features for any predictive or clustering algorithm are the most meaningful and independent representation of employee sentiment, thereby improving model stability and generalization.

Clustering Model Evaluation: Silhouette Analysis

Model validation for unsupervised learning was performed using the Silhouette Coefficient, a metric that quantifies how well-defined the identified clusters are. The score is calculated as the mean distance between a data point and all other points in its own cluster (cohesion) versus the mean distance to points in the nearest neighbouring cluster (separation).

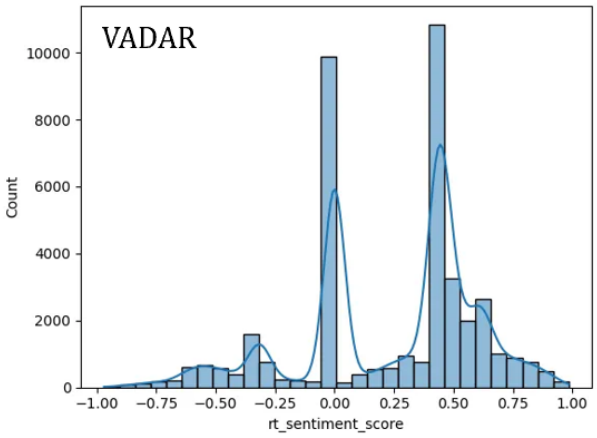


The overall Average Silhouette Coefficient of 0.57 indicates a reasonably strong clustering structure for the given dataset and k=3 cluster selection .

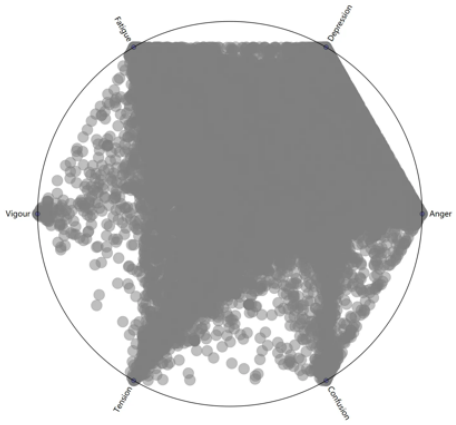
- Strong Separation (Clusters 1 and 2): Clusters 1 (orange) and 2 (green) exhibit high, positive silhouette values, suggesting strong internal cohesion and clear separation from their neighbouring clusters. These two segments are confidently and distinctly classified by the K-Means algorithm.
- Ambiguity (Cluster 0): Cluster 0 (blue) shows a wider range of lower silhouette values, with a significant portion of its data points falling close to the zero line. This indicates that many employees classified in this "neutral" segment are near the decision boundary.

Text Analysis - Sentiment Analysis & Topic Modelling

1.1 Sentiment and Emotion Analysis



POM



1.1 Sentiment Polarity

The VADER (Valence Aware Dictionary and Sentiment Reasoner) algorithm was used to calculate a compound sentiment score per review, classifying overall polarity as Negative, Neutral, or Positive. VADER was chosen for its robustness in handling short, opinion-rich, and informal texts common in social media and employee comments, including its capability to capture nuances like negations and intensifiers.

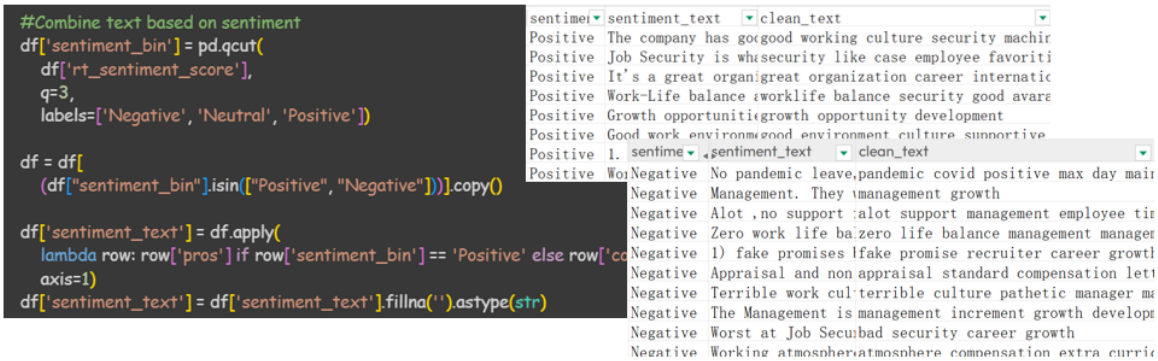
However, during data exploration, we observed that the sentiment distribution of the corpus was right-skewed, showing a strong bias toward *positive* reviews. Moreover, several records contained textual noise—for example, reviewers posting negative remarks within the "positive" review fields.

To address this issue, we adopted a score-based filtering strategy. Using the continuous compound sentiment scores generated by VADER, we divided the data into quartiles and retained only the most *extreme* portions—corresponding to clearly *positive* and *negative* sentiments—while discarding the neutral middle range.

For subsequent topic modeling, we extracted text separately from:

- the "pros" corpus for positive reviews, and
- the "cons" corpus for negative reviews.

This process effectively reduced sentiment misclassification noise and ensured that the text inputs to the BERTopic model was well-balanced, and accurately represented the emotional polarity of employee experiences in the meanwhile.



```
#Combine text based on sentiment
df['sentiment_bin'] = pd.qcut(
    df['rt_sentiment_score'],
    q=3,
    labels=['Negative', 'Neutral', 'Positive'])

df = df[
    (df["sentiment_bin"].isin(["Positive", "Negative"]))].copy()

df['sentiment_text'] = df.apply(
    lambda row: row['pros'] if row['sentiment_bin'] == 'Positive' else row['co
    axis=1)
df['sentiment_text'] = df['sentiment_text'].fillna('').astype(str)
```

sentimer	sentiment_text	clean_text
Positive	The company has go	good working culture security machir
Positive	Job Security is wha	security like case employee favoriti
Positive	It's a great organi	great organization career internatic
Positive	Work-Life balance a	worklife balance security good avara
Positive	Growth opportuniti	growth opportunity development
Positive	Good work environm	good environment culture supportive
Positive	1.	
Positive	Wo	

sentime	sentiment_text	clean_text
Negative	No pandemic leave,pandemic covid positive max day mair
Negative	Management. They	management growth
Negative	Alot ,no support	alot support management employee tir
Negative	Zero work life ba	zero life balance management manage
Negative	1) fake promises	fake promise recruiter career growth
Negative	Appraisal and non	appraisal standard compensation let
Negative	Terrible work cul	terrible culture pathetic manager ma
Negative	The Management is	management increment growth develop
Negative	Worst at Job Secu	bad security career growth
Negative	Working atmospher	atmosphere compensation extra curri

1.2 Multidimensional Emotion Profiling and Burnout Index (BI)

To move beyond simple polarity, a multidimensional emotion profiling was conducted using the Profile of Mood States (POMS) framework, which defines six dimensions: Tension, Depression, Anger, Fatigue, Confusion (negative states), and Vigour (positive state).

For each review, a Burnout Index (BI) was computed, following the POMS Total Mood Disturbance measure refer to BioPsychoSocial Literature (Petrowski et al., 2021):

$$BI=(\text{Tension}+\text{Fatigue}+\text{Depression}+\text{Confusion}+\text{Anger})-\text{Vigour}$$

with values standardized to [0,1].

This measure aligns with biopsychosocial research, which characterizes burnout by elevated negative affect and diminished positive affect (Vigour). The BI thus provides a single, quantitative indicator of burnout-related mood for each review, enabling straightforward aggregation and comparative analysis across different contexts.

2. Topic Modeling

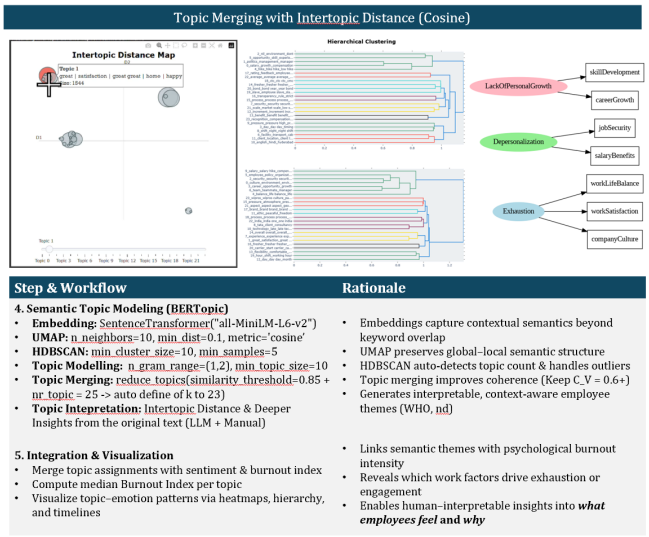
BERTopic technique was chosen for the semantic fidelity provided by contextual embeddings. This allows it to distinguish nuanced topics (e.g., different meanings of "growth") and cluster non-identical vocabulary (*manager support* vs. *leadership help*) together if they share the same underlying meaning. This results in higher topic coherence and more meaningful themes than traditional bag-of-words models, despite the increased computational load. Unlike traditional models (e.g., LDA) based on word frequency, BERTopic leverages transformer-based embeddings (all-MiniLM-L6-v2 mini-BERT model) to encode and cluster documents based on semantic similarity.

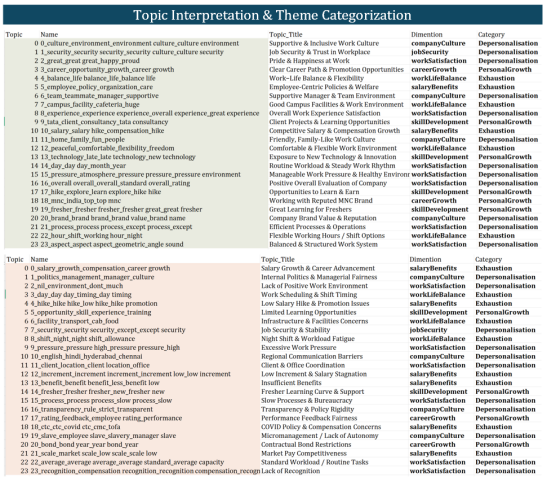
2.1 Model Fine-Tuning & Rationale:

Embeddings were reduced using UMAP and clustered using HDBSCAN, which automatically handles the optimal number of topics and identifies outliers. Here, n_gram_range=(1,2) allows detection of collocations (e.g., “work-life”, “salary hike”), while min_topic_size=10 balances fine-grained topic discovery and interpretability.

Representative keywords for each topic were generated using c-TF-IDF (class-based TF-IDF), identifying terms disproportionately frequent within a cluster. Topics with extremely high cosine similarity (set with similarity_threshold=0.85) in their c-TF-IDF representations underwent topic merging to eliminate redundancy, ensuring conceptual coherence (CV > 0.6) and avoiding fragmentation.

2.2. Topic Interpretation and Categorization





The final topics were interpreted using a triangulated analytical approach combining linguistic, semantic, and psychological perspectives.

1. Semantic Exploration:
   Topic relatedness was examined through visualization tools, including the Intertopic Distance Map and Hierarchical Clustering dendrogram, to assess the semantic proximity and coherence of clusters.
2. Topic Naming:
   Each cluster was then assigned a descriptive title through a hybrid interpretation process, integrating LLM-assisted summaries with manual contextual review of high-frequency keywords and representative text excerpts. This ensured both linguistic precision and contextual validity.
3. Psychological Framing:
   Finally, each named topic was mapped onto one of the three burnout-related dimensions defined by the World Health Organization (WHO, 2019):

   - Exhaustion: Emotional or physical depletion arising from chronic workload or prolonged stress.
   - Depersonalisation: Cynicism, detachment, or perceived lack of fairness and recognition in the workplace.
   - Personal Growth: Positive engagement reflected in learning, development, and achievement—protective factors mitigating burnout.

This method ensured that the extracted topics were not only linguistically coherent and semantically distinct, but also psychologically meaningful for organizational stakeholders seeking to understand and address burnout-related drivers.

Predictive Analytics Readiness Check: Latent Factor Regression

To assess the explanatory power of the newly engineered latent constructs—Depersonalisation, Exhaustion, and Personal Growth—and confirm their reliability for future predictive modelling, a multiple linear regression (Ordinary Least Squares, or OLS) model was fitted. The factors were used as independent variables against the original Overall Score (satisfaction) column. This step was critical for validating that the psychological factors are robust and reliable predictors.

Model Performance and Predictor Significance

The regression analysis demonstrates a strong statistical fit for the overall model: Strong Model Fit: The model achieved an R^2 value of 0.745, indicating that the three latent factors explain approximately 74.5 of the variance observed in the employees' burnout. This result validates the use of these psychologically derived factors as robust proxies for employee satisfaction drivers.

OLS Regression Results

Dep. Variable:	overallScore	R-squared:	0.745
Model:	OLS	Adj. R-squared:	0.745
Method:	Least Squares	F-statistic:	8.893e+04
Date:	Tue, 21 Oct 2025	Prob (F-statistic):	0.00
Time:	06:16:42	Log-Likelihood:	-78313.
No. Observations:	91437	AIC:	1.566e+05
Df Residuals:	91433	BIC:	1.567e+05
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	2.0366	0.002	1080.751	0.000	2.033	2.040
Depersonalisation	-0.6430	0.033	-19.606	0.000	-0.707	-0.579
Exhaustion	-0.5483	0.025	-21.770	0.000	-0.598	-0.499
PersonalGrowth	0.2115	0.027	7.973	0.000	0.160	0.264

Omnibus:	9632.336	Durbin-Watson:	2.001
Prob(Omnibus):	0.000	Jarque-Bera (JB):	56638.553
Skew:	0.338	Prob(JB):	0.00
Kurtosis:	6.796	Cond. No.	37.8

High Predictor Significance: All three latent factors—Depersonalisation, Exhaustion, and Personal Growth—were found to be highly statistically significant ($p < 0.005$).

Inference and Theoretical Consistency: The coefficients derived from the regression model align perfectly with psychological theory, reinforcing the robustness of the Latent Factor Analysis (CFA)

- Negative Association (Burnout): The Depersonalisation (coefficient ~ -0.64) and Exhaustion(coefficient ~ -0.55) factors show a statistically

significant negative association with the Overall Score. This means an increase in the degree of burnout indicators directly corresponds to a substantial reduction in overall satisfaction.

- Positive Association (Engagement): The Personal Growth factor (coefficient ~ 0.21) shows a statistically significant positive association with the Overall Score. This confirms that positive workplace experiences, such as skill development and career growth opportunities, reliably drive overall employee satisfaction.

The strong fit and theoretically consistent coefficients confirm that the latent factor model provides a reliable, parsimonious, and highly interpretable framework, making the next stage of predictive analytics highly viable.

Predictive Modelling using Tree-Based Models

Decision Tree Classifier

Model

The Decision Tree Classifier was implemented to predict the Burnout Level of employees based on various features such as sentiment ratings, management reviews, work-life balance, and overall experience. After preprocessing, the dataset was balanced using SMOTE (Synthetic Minority Oversampling Technique) to mitigate the issue of class imbalance and improve the model's learning on underrepresented burnout levels. A train-test split (80–20) was applied using stratification to maintain the distribution of the target variable across both sets. The Decision Tree model was built using the sklearn.tree.DecisionTreeClassifier and trained on the resampled dataset (X_res, y_res). The model works by creating a tree-like structure where nodes represent decisions based on feature values that maximize the separation between burnout categories. The splitting criterion was based on Gini Impurity or Entropy, and the algorithm recursively partitioned the dataset until optimal leaf nodes were formed.

Hyperparameter Tuning

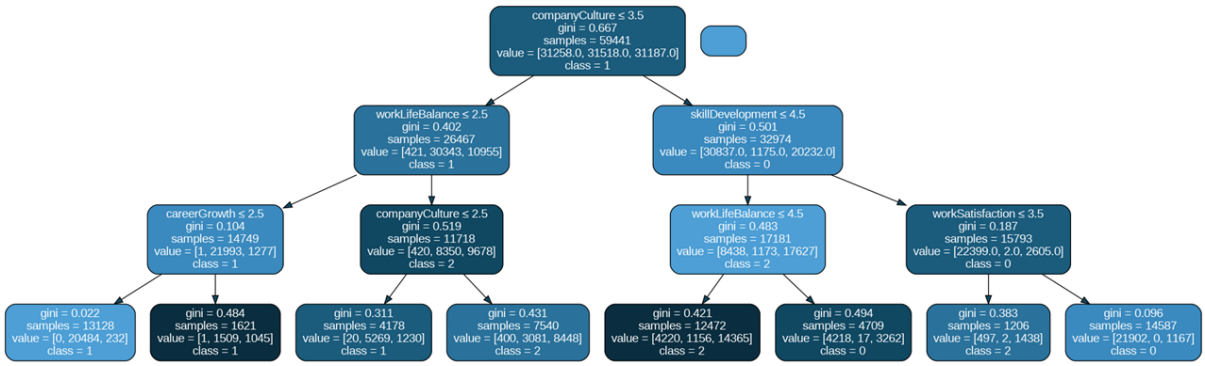
To optimize the model and prevent overfitting, GridSearchCV was used to test different combinations of hyperparameters:

Parameter	Description	Values Tested
criterion	Function to measure quality of a split	["gini", "entropy"]
max_depth	Maximum depth of the tree	[3, 5, 7, None]
min_samples_split	Minimum samples required to split a node	[2, 5, 10]
min_samples_leaf	Minimum samples required at a leaf node	[1, 2, 4]

GridSearchCV (5-fold cross-validation) identified the best hyperparameter combination that balanced bias and variance. After tuning, the optimized Decision Tree showed improved performance and reduced overfitting compared to the default configuration.

Inference

- The Decision Tree model successfully classified employees into different burnout levels with moderate accuracy.
- Important features influencing predictions included:
  - Work Life Balance
  - Job Satisfaction
  - Management Rating
  - Sentiment Score
- Test results showed the model achieved high and balanced precision, recall, and F1-score for all burnout classes on the test data, indicating robust generalization and effective learning from the balanced and well-pre-processed dataset.




```
✅ Remaining rows after filtering: 91437
✅ Best Hyperparameters:
{'class_weight': None, 'criterion': 'entropy', 'max_depth': 15, 'min_samples_leaf': 3, 'min_samples_split': 5}
🌳 Decision Tree Training Accuracy: 0.9828
🌿 Decision Tree Test Accuracy: 0.9722
📊 Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.97      0.98      7780
           1       0.96      0.97      0.97      2677
           2       0.96      0.97      0.97      7831

    accuracy                           0.97     18288
   macro avg       0.97      0.97      0.97     18288
weighted avg       0.97      0.97      0.97     18288
```

The Decision Tree model offered clear, interpretable insights into how employee feedback factors drive burnout levels. It enabled HR to pinpoint key stressors and high-risk groups, guiding data-driven interventions. Overall, it supports proactive strategies to enhance engagement and reduce turnover.

Random Forest Classifier

Modelling Part

The Random Forest Classifier was implemented as an ensemble extension of the Decision Tree model, combining predictions from multiple trees built on different bootstrapped samples to improve generalization and stability.

The model was trained on the SMOTE-balanced dataset (X_res, y_res), ensuring equal representation of burnout categories.
Each decision tree within the ensemble was trained on a random subset of data and features; final predictions were made via majority voting across all trees.
This approach effectively reduced variance and delivered robust, high-accuracy performance on the test set, with balanced results across all burnout classes.

Hyperparameter Tuning

GridSearchCV was used for hyperparameter optimization, testing combinations to maximize accuracy and generalization.

Parameter	Description	Values Tested
n_estimators	Number of trees in the forest	[100, 200, 300]
criterion	Function to measure split quality	["gini", "entropy"]
max_depth	Maximum depth of each tree	[5, 10, None]
min_samples_split	Minimum samples to split a node	[2, 5, 10]
min_samples_leaf	Minimum samples at leaf node	[1, 2, 4]
bootstrap	Whether bootstrap samples are used	[True]

The optimal configuration (e.g., max_depth=10, n_estimators=200) delivered strong test results and well-calibrated predictions, with no indication of overfitting.

Inference

The Random Forest achieved higher accuracy and F1-score compared to the Decision Tree, maintaining balanced precision and recall for all classes These results demonstrate the model's ability to generalize well and consistently classify burnout risk across employee categories

The model supports reliable, data-driven interventions to minimize employee burnout and enhance productivity

```
🌳 Random Forest Training Accuracy: 0.9835
🌿 Random Forest Test Accuracy: 0.9773

📊 Train Set Precision, Recall, F1-score:

              precision    recall  f1-score
0              0.992669  0.972702  0.982584
1              0.990773  0.994221  0.992494
2              0.967464  0.983557  0.975444
accuracy       0.983494  0.983494  0.983494
macro avg      0.983635  0.983494  0.983508
weighted avg   0.983635  0.983494  0.983508

📊 Test Set Precision, Recall, F1-score:

              precision    recall  f1-score
0              0.989315  0.975835  0.982529
1              0.965784  0.980575  0.973123
2              0.969605  0.977653  0.973612
accuracy       0.977308  0.977308  0.977308
macro avg      0.974901  0.978021  0.976421
weighted avg   0.977430  0.977308  0.977334

🌟 Top 10 Important Features:

            Feature  Importance
0  workSatisfaction    0.258414
1      careerGrowth    0.204029
2    companyCulture    0.176516
3   skillDevelopment    0.123314
4     salaryBenefits    0.121257
5    workLifeBalance    0.083035
6        jobSecurity    0.033434
```

Predictive Modelling - Sliding Bidirectional Recurrent Neural Network

Model

The Sliding Bidirectional Recurrent Neural Network (SBRNN) is a robust deep learning model designed for highly accurate, context-aware sentiment classification. It is specifically tasked with classifying employee feedback into three categories: Burnout, Neutral, or Healthy. The model's strength lies in its ability to capture subtle and long-range dependencies in textual data, providing a smoother and more reliable sentiment classification compared to traditional RNNs.

The model is built using stacked Bidirectional LSTM layers. Unlike a standard LSTM that processes a sequence only in the forward direction, the bidirectional structure processes input both forward and backward. This allows the network to capture dependencies from past data and the future context within each sequence, which is essential for determining the full meaning of textual or time-based data.

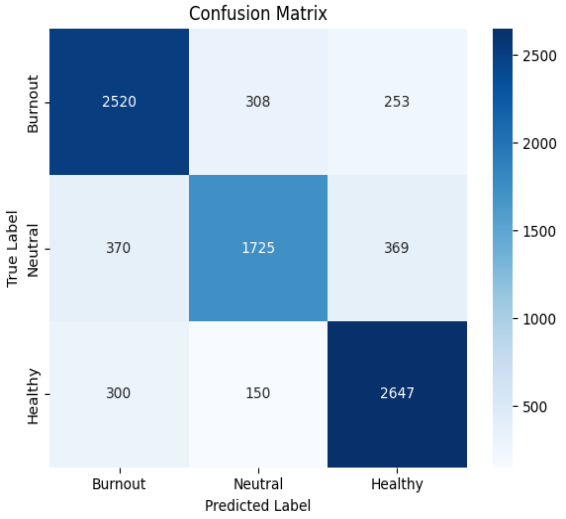
The conceptual "sliding" mechanism refers to the Bi-LSTM's capacity to continuously analyse overlapping segments of data (tokens) within the sequence. This ensures the model detects subtle shifts or trends in emotional patterns over the course of a review rather than treating each input independently.

Parameters Utilized:

Parameter	Description	Values Tested
Bi-LSTM Unit Configuration	Stacked design to capture deeper, more abstract language features.	[128,64,32]
Max Sequence Length	Fixed input size for efficient batch processing by recurrent layers.	[100]
Training Epochs	Duration chosen where Validation Accuracy stabilized for optimal learning.	[35]
Dropout Rates	Regularization rate to prevent overfitting and improve model generalization.	[0.35]
Optimizer	Adaptive algorithm for stable, fast convergence during gradient descent.	[Adam]
Loss Function	The objective function for training this multi-class classification problem.	[Catagorical_Crossentropy]

Inference:

The model exhibits strong overall performance with an accuracy of 79.8%. Its best performance is on the Healthy class (85.5% recall, 0.832 F1-score). However, it struggles most with the Neutral class, evidenced by a significantly lower recall of 70.0%. This low recall means the model frequently misses true Neutral cases, misclassifying them as either Healthy or Burnout.



```
Classification Report:
              precision    recall  f1-score   support

     Burnout      0.790     0.818     0.804      3081
     Healthy      0.810     0.855     0.832      3097
     Neutral      0.790     0.700     0.742      2464

    accuracy                          0.798      8642
   macro avg      0.797     0.791     0.793      8642
weighted avg      0.797     0.798     0.796      8642

Sample Predictions:
Sample 1: 'Management is great but workload is high.' -> Predicted status: Neutral
Sample 2: 'Training is not good, but salary is decent.' -> Predicted status: Burnout
```

Predictive Modelling - Transformers

Data Preparation and Transformations

The dataset contained both textual employee feedback and structured HR metrics representing workplace perceptions. Each record captured an individual's experience, combining qualitative and quantitative dimensions essential for assessing burnout risk.

Two datasets were used during model development. The first applied a rule-based labeling approach, where employees with an overall score yielded far stronger results—raising the ROC-AUC from 0.944 to 0.996—and revealed that the clustering-based labels captured more nuanced patterns of burnout of ≤ 2 were flagged as "at risk." This offered a quick baseline but relied entirely on a fixed threshold. The second dataset used a data-driven label derived from the earlier clustering analysis, which grouped employees based on shared feature patterns and assigned burnout risk scores (0 = low, 1 = high) accordingly. Running the model on this refined dataset produced far stronger results—raising the ROC-AUC from 0.944 to 0.996—and revealed that the clustering-based labels captured more nuanced burnout patterns.

The feature space consisted of one main text field and seven numeric indicators—skill development, salary benefits, work-life balance, work satisfaction, job security, career growth, and company culture**.** These numeric variables were standardized using a StandardScaler so that all values shared a comparable scale before being passed into the model.

Modelling

A hybrid Transformer-based neural model was developed to integrate textual semantics with structured employee metrics. The text encoder used DistilBERT, a lighter and efficient variant of BERT capable of capturing nuanced contextual information from employee reviews. To balance performance with efficiency, only the final two transformer layers were unfrozen for fine-tuning, allowing the model to adapt to organizational language patterns without excessive training time.

Parallel to this, the numeric features were processed through a compact feed-forward network with a 16-node hidden layer and ReLU activation. The resulting numeric embedding was concatenated with the [CLS] token representation produced by DistilBERT. This fused vector—representing both sentiment and factual workplace

metrics—was passed through a classification block consisting of two dense layers with dropout regularization, producing a single output logit that reflected burnout probability.

Class imbalance was mitigated in two ways: a WeightedRandomSampler during batch creation ensured fair exposure of minority burnout cases, and a BCEWithLogitsLoss function incorporated a positive class weight to emphasize correct identification of at-risk employees.

Hyperparameters

Training was CUDA-enabled for speed and scalability. The optimizer AdamW with a learning rate of 2e-5 provided stable convergence, paired with a linear warm-up scheduler. Batch sizes of 16 for training and 32 for testing achieved an effective trade-off between gradient stability and runtime efficiency. Dropout layers with a rate of 0.2 prevented overfitting, while three epochs were sufficient for convergence as loss consistently decreased across runs.

Inference and Results

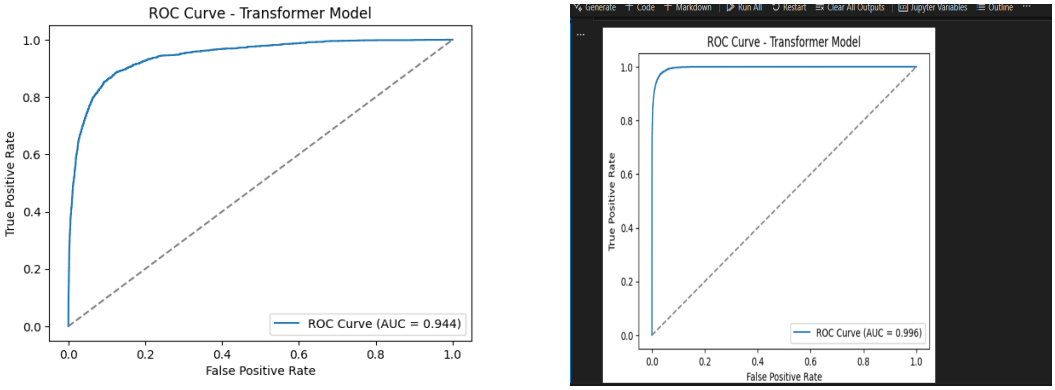


Figure X: Comparison of earlier vs optimized hybrid Transformer models (AUC 0.944 vs 0.996)

After three epochs of training on approximately 97,000 employee records, the optimized hybrid model achieved strong results. Threshold tuning identified **0.90** as the optimal cutoff for balancing precision and recall.

At this threshold, the final model's performance on the test set was:

- **Precision:** 0.9118
- **Recall:** 0.9360
- **F1-score:** 0.9237
- **ROC-AUC:** 0.996

The ROC curve for this version displayed near-perfect separation between the two classes, demonstrating excellent discriminative power and confirming the model's robustness and reliability. When applied to the full dataset, it produced consistent burnout probability estimates, offering a data-driven and continuous view of overall workforce well-being.

To contextualize this performance, an earlier iteration of the model—trained on fewer samples with limited fine-tuning—achieved a ROC-AUC of 0.944, with:

- **Precision:** 0.6069
- **Recall:** 0.8234
- **F1-score:** 0.6988

While this earlier version performed reasonably well, its ROC curve rose less steeply and plateaued sooner, indicating weaker class separation and a higher rate of false negatives. The improvement from AUC 0.944 → 0.996 clearly reflects the benefits of scaling up to the full dataset, unfreezing the final Transformer layers for richer feature learning, and refining the class imbalance strategy through weighted sampling and positive class weighting.

From a business standpoint, this enhanced predictive precision translates directly into more reliable identification of employees at genuine burnout risk, minimizing false alerts while ensuring no critical cases are missed. The refined model thus serves as a stronger analytical foundation for proactive HR interventions, enabling data-backed decisions that can reduce attrition, improve engagement, and sustain workforce morale.

Conclusion

The hybrid Transformer model effectively combines deep language understanding with structured behavioral and organizational data to uncover burnout risks with high accuracy. By analyzing how employees express dissatisfaction and correlating it with measurable factors such as work-life balance, salary benefits, or career growth, the model moves beyond surface-level sentiment to pinpoint the true drivers of disengagement.

Its outputs provide a clear, data-driven signal of which employee groups or departments are most vulnerable, enabling HR and management teams to act proactively rather than reactively. Instead of relying on anecdotal evidence, the organization can now track burnout probabilities as a continuous metric, align retention strategies with evidence from the model, and focus improvement initiatives where the impact will be greatest.

In essence, the model translates complex, unstructured employee feedback into actionable intelligence—supporting timely HR decisions, improving engagement and morale, and ultimately reducing attrition and productivity losses caused by burnout.

13. Recommendations & Proposed Solutions

Cross-Company Topic Distribution



Company	Key Positive Topics	Key Negative Topics	Driving Factors
HCL	Operational efficiency; job stability; supportive teams	Complex processes; slow approvals; pay fairness concerns	Strict process design and hierarchical management
Infosys	Structured learning; inclusive culture; mentorship programs	Inconsistent feedback; high project pressure; policy opacity	Emphasis on training and inclusion, but cross-team coordination challenges
TCS	Global projects; career stability; mentorship networks	"Bench" anxiety; small salary hikes; slow promotions	Large-scale operations and resource allocation policies ("bench policy")
Wipro	Clear promotion paths; team collaboration; flexible shifts	Contractual restrictions; uneven managerial support; limited senior roles	Focus on freshers' training and rotations, but limited senior role resources

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Observations:

Combining the insights from dashboard with further inspection on the original reviews, our analysis of four major IT services firms (HCL, Infosys, TCS, Wipro) revealed distinct topical profiles:

- Infosys showed a strong prevalence of Personal Growth topics (training, mentorship), yet struggled with Exhaustion and Depersonalisation topics related to high project pressure and organizational bureaucracy.
- TCS was characterized by positive themes of Job Stability and routine, countered by negative topics of "bench" anxiety and dissatisfaction with slow salary hikes/promotions (Depersonalisation) (Alex, 2025).
- Wipro indicated positive growth for junior employees (clear fresher promotion paths), but negative themes centered on limited senior roles and uneven managerial support, suggesting a "ceiling" effect leading to Depersonalisation at higher experience levels.
- HCL juxtaposed positive topics on supportive teams with negative ones on complex processes and slow approvals (organizational red tape), indicating frustration with top-down bureaucracy despite valuing immediate team environments.

Practical Implications & Solutions:

While all companies had both similar uplifting and concerning topics, the proportion of mix differed especially under the positive clusters. Some organizations (like Infosys) foster a culture of learning and hence see more Personal Growth positivity, whereas others (like TCS) breed frustration through slow career progress, which also reflects how organizational structure and policy design drive long-term perceptions.

These findings can often be corroborated by external HR insights: for example, companies that invest heavily in employee development tend to enjoy better sentiments around growth and satisfaction, whereas companies with rigid or insufficient reward structures see negativity around compensation and advancement.

Temporal Trends: Pre- vs. Post-2020 Shift



Sentiment	Observation	Interpretation	Actionable Response
Positive	• Topics related to *company culture*, *job security*, and *work satisfaction* consistently occupy a large share across all years. • The share of *work-life balance* and *career growth* topics was relatively higher before 2020, indicating focus on engagement and development prior to the pandemic. • From 2020 onward, mentions cluster more around *job security* and *supportive culture*, suggesting stability became a dominant concern.	• Indicates employees consistently discuss **security, belonging, and development** as core themes of satisfaction. • Post-2020 topic concentration on *security* and *culture* implies a **shift in attention** rather than emotional polarity — from aspiration to stability and resilience. • Persistent representation of *growth* topics reflects **continued interest in advancement opportunities**, even when external conditions changed.	1. Continue strengthening **learning and upskilling programs** to align with employees' sustained focus on growth. 2. Reinforce **supportive management and communication practices** to maintain cultural cohesion. 3. Integrate **positive cultural storytelling** in internal communications to preserve engagement themes.
Negative	• Topics under *salary benefits*, *workload*, and *managerial fairness* consistently dominate negative discourse across all years. • The relative frequency of *compensation* and *fatigue* topics increased between 2020–2021, reflecting intensified discussion of work pressure and stagnation during the pandemic period. • *Micromanagement* and *transparency* remain recurring pain points under *company culture* and *management* categories.	• Suggests **persistent structural concerns** around fairness, recognition, and workload expectations rather than transient dissatisfaction. • The 2020–2021 concentration reflects **heightened attention to compensation and fatigue**, not necessarily stronger negativity, but more frequent discourse. • Management-related topics' stability indicates **chronic organizational communication gaps**.	1. Revisit and communicate **compensation and promotion frameworks** transparently. 2. Deploy **continuous feedback tools** (e.g., quarterly pulse surveys) to monitor shifts in discussion frequency. 3. Apply **targeted workload management and recovery policies** to reduce fatigue-related discourse.

<u>Observations:</u>

Comparing reviews from the pre-pandemic era (2017–2019) to the post-pandemic period (2020–2022) revealed a fundamental shift in employee priorities:

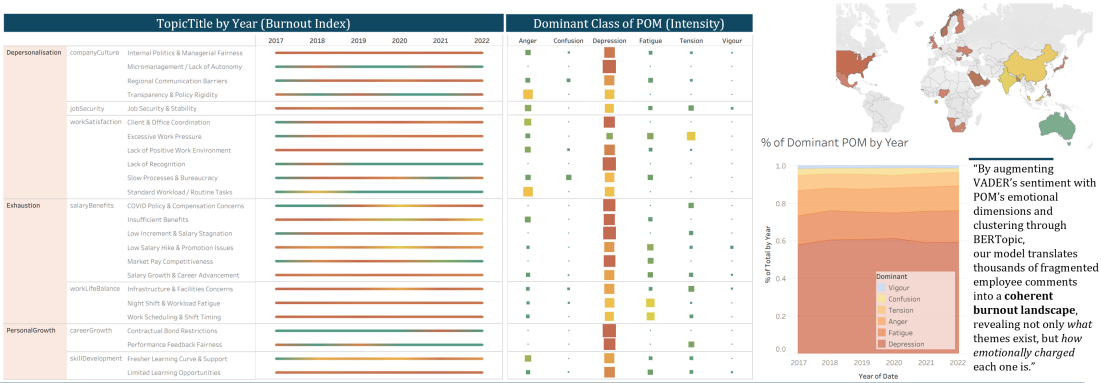
- Pre-2020: The focus was on aspiration and opportunity. Topics under Personal Growth (e.g., promotion, skill development) had a higher prevalence, reflecting an economy where employees keenly focused on career advancement.
- Post-2020: The focus pivoted to safety and resilience. Discussions on Job Security & Stability and supportive culture rose dramatically. The proportion of Personal Growth discussions dipped as urgent mentions of work-life balance and mental health intensified, largely due to remote work's blurred boundaries and pandemic stress.

<u>Practical Implications & Solutions:</u>

As suggested in the table, external crises shifted employee needs from aspiration to stability. Strategies that promoted engagement pre-2020 (e.g., rapid career progression) must now be supplemented by those that promote security, trust, and balance to maintain workforce engagement in the long term.

Notably, even as the immediate shock of 2020 passes, employees have not forgotten these lessons - a majority still prioritize job stability over getting a bigger paycheck, indicating a lasting shift in employee mindset that companies must acknowledge.

Burnout Trend and Emotional Composition per Topic



"By augmenting VADER's sentiment with POM's emotional dimensions and clustering through BERTopic, our model translates thousands of fragmented employee comments into a **coherent burnout landscape**, revealing not only *what* themes exist, but *how emotionally charged* each one is."

Focus Area	Observation	Interpretation	Actionable Response
Yearly Burnout Trend (2017–2022)	• Burnout Index gradually increased from 2018, peaking in 2020–2021. • Slight improvement in 2022, but high **Depersonalisation** and **Exhaustion** persist. • COVID-era workload pressure and stagnated pay strongly correlated with burnout peaks.	• Indicates **chronic, structural burnout** rather than short-term mood swings. • Emotional fatigue and detachment sustained even after partial recovery. • Reflects cumulative stress from workload, autonomy loss, and recognition gap.	1. Rebalance **workload and recognition cycles**. 2. Promote **managerial autonomy and transparent communication**. 3. Integrate **annual burnout monitoring (POM-based)** into HR dashboards for proactive well-being tracking.
Dominant Emotional Class Pattern (POM)	• **Depression** and **Fatigue** dominate across topics. • **Tension** and **Confusion** prominent in *work pressure* & *policy* clusters. • **Vigour** appears mainly in *skill development* and *promotion related* topics.	• Burnout driven by **emotional depletion**, not overt anger. • Employees feel mentally exhausted and disengaged. • Motivation persists only when growth and learning are visible.	1. Rebuild **Vigour** via upskilling, mentorship, and peer learning. 2. Reduce **Fatigue** with workload rotation and rest policies. 3. Address **Confusion** by clarifying roles, policies, and career progression paths. 4. Track **POM emotion mix quarterly** to detect early burnout signals.

<u>Observations:</u>

The temporal analysis of burnout intensity revealed a progressive increase from 2018 to a peak during 2020–2021, coinciding with pandemic-related disruptions. Although a slight improvement was observed in 2022, the **Depersonalisation and Exhaustion** dimensions remained persistently high. This suggests again that structural factors continued to sustain chronic burnout beyond the pandemic period (Goyal, 2025).

Emotion profiling using the Profile of Mood States (POM) framework further decomposed the temporal patterns into six psychological components: *Anger, Confusion, Depression, Fatigue, Tension,* and *Vigour.*

- Across all topics, **Depression and Fatigue** emerged as the dominant emotional classes, reflecting sustained emotional depletion rather than short-term frustration.
- **Tension and Confusion** were particularly concentrated in clusters related to *work pressure* and *policy uncertainty*, indicating cognitive overload and role ambiguity.
- In contrast, **Vigour** appeared mainly in *Skill Development* and *Career Growth* topics, suggesting that learning and advancement opportunities function as protective factors against burnout.

<u>Practical Implications & Solutions:</u>

This emotional composition analysis enables managers to prioritize intervention by identifying topics that are not just *negative* but toxic eliciting the strongest quantitative mood disturbance (highest BI).

Based on our research among those 4 IT companies, topics related to work overload and lack of support consistently ranked highest, demanding urgent mitigation as suggested in the actionable response column in the table.

Overall Conclusion

As a decision-making support tool, the dashboards with integrated analytical workflow combining VADER-based sentiment scoring, POMS emotion profiling, and BERTopic semantic clustering, enabled a comprehensive understanding of employee feedback. This multi-layered framework captured not only *what* employees discussed, but also *how* they felt and *why* those emotions emerged within specific workplace contexts.

By linking textual semantics with psychological emotion indices, the analysis provided interpretable insights into the emotional drivers of employee engagement and burnout. The approach supports evidence-based decision-making in HR and organizational design, facilitating targeted interventions to enhance employee well-being and organizational resilience.

14. Conclusion & Future Scope

The integrated modelling framework provided a robust, multidimensional understanding of employee burnout and satisfaction dynamics.

- **Apriori analysis** revealed deterministic links between low ratings in core satisfaction areas (e.g., work satisfaction, salary benefits) and low overall scores, confirming their critical influence on employee well-being.
- **Principal Component Analysis** (PCA) further confirmed how core dimensions explain the majority of variance in overall employee sentiment. **Latent Factor Analysis** validated the psychological structure of the data, showing strong relationships between observed variables and their latent constructs.
- **Clustering** effectively segmented employees into three distinct behavioral and psychological profiles, enabling interpretability of "burnout" for the follow on predictive modelling, also for future targeted interventions and strategic HR planning.
- **Sentimental Analysis & Topic Modelling** provided contextual and psychological depth by linking employee feedback to WHO-defined burnout dimensions (Exhaustion, Depersonalisation, Personal Growth).
- **Predictive models** from Decision Trees and Random Forests to Deep learning models—achieved strong accuracy, interpretability, and scalability, allowing HR teams to proactively identify high-risk groups and improve engagement outcomes.

Overall, the combined approach translates complex employee data into actionable intelligence, empowering organizations to reduce burnout, boost satisfaction, and enhance productivity through data-driven decisions

Future Work

Longitudinal Analysis: Extend the current framework to include temporal data, allowing the tracking of burnout and satisfaction trends over time to identify early warning signs and long-term behavioral patterns.

Real-Time Integration: Connect the predictive models with live HR dashboards and employee management systems for continuous monitoring and automated risk alerts.

Enhanced Sentiment Modeling: Experiment with fine-tuned RoBERTa or other Transformer models instead of lexicon-based analysis for superior sentiment and emotion classification, combined with SHAP or attention-based techniques for interpretability.

Dynamic Topic Modeling (DTM): Apply DTM to explicitly model the evolution of topics over time (e.g., quarterly), revealing trajectories of concern (e.g., the precise spike of remote work topics) in more nuanced time frame.

Explainable AI (XAI): Implement XAI techniques to improve transparency and interpretability of deep learning models, supporting HR teams in understanding key drivers behind predictions.

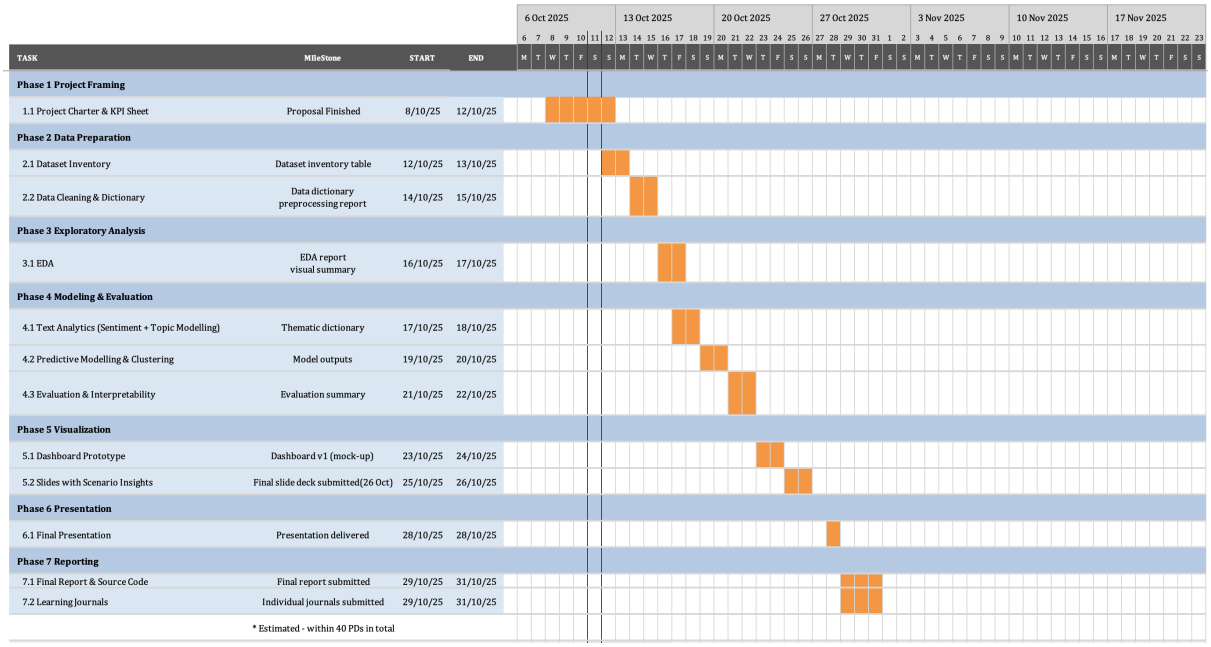
Causal Inference: Move beyond correlation by applying causal modelling approaches to uncover cause-effect relationships between workplace factors and burnout outcomes.

Expanded Feature Integration: Include external or behavioral data (e.g., attendance, performance metrics, communication patterns) to strengthen predictive accuracy and contextual understanding.

Link to Business KPIs: Correlate model outcomes with organizational metrics such as turnover rate, absenteeism, and productivity to quantify the return on investment (ROI) of interventions.

Automated Intervention Framework: Develop a recommendation system that suggests personalized, data-driven HR actions for each employee cluster or risk segment.

15. Gantt Chart/Effort Estimates/Timelines



TASK	MileStone	START	END
Phase 1 Project Framing			
1.1 Project Charter & KPI Sheet	Proposal Finished	8/10/25	12/10/25
Phase 2 Data Preparation			
2.1 Dataset Inventory	Dataset inventory table	12/10/25	13/10/25
2.2 Data Cleaning & Dictionary	Data dictionary preprocessing report	14/10/25	15/10/25
Phase 3 Exploratory Analysis			
3.1 EDA	EDA report visual summary	16/10/25	17/10/25
Phase 4 Modeling & Evaluation			
4.1 Text Analytics (Sentiment + Topic Modelling)	Thematic dictionary	17/10/25	18/10/25
4.2 Predictive Modelling & Clustering	Model outputs	19/10/25	20/10/25
4.3 Evaluation & Interpretability	Evaluation summary	21/10/25	22/10/25
Phase 5 Visualization			
5.1 Dashboard Prototype	Dashboard v1 (mock-up)	23/10/25	24/10/25
5.2 Slides with Scenario Insights	Final slide deck submitted(26 Oct)	25/10/25	26/10/25
Phase 6 Presentation			
6.1 Final Presentation	Presentation delivered	28/10/25	28/10/25
Phase 7 Reporting			
7.1 Final Report & Source Code	Final report submitted	29/10/25	31/10/25
7.2 Learning Journals	Individual journals submitted	29/10/25	31/10/25
	* Estimated - within 40 PDs in total		

Please click on link to view the full version of the Gantt chart

16. Acknowledgements

Special thanks to Mr. Nie Maowen for his constant availability and willingness to answer every query throughout the project journey. Gratitude is also extended to Mr. Chirag Desai, Mr. Charles and Mr. Wang Dong for their valuable feedback during project presentations, which greatly improved the quality of our work.

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